



RKDF University, Bhopal

Open Distance Learning (ODL)

Material

Faculty of Commerce

Semester -I

Subject- Data processing Software

Course outcomes:-

1. To understand the basic concept of various Applications of software.
2. To gain knowledge of MS Word, Excel, Access and Power point.
3. To apply acquired knowledge in office automation tasks.
4. To study various methods of formatting of documentation and use of spread sheets.
5. To develop and enhance presentation skills using power point.

Syllabus

Units	Topic	Duration (In Hours)	Marks
I	MS Windows: Introduction to Ms Windows; Features of Windows; Various version of Windows & their use; Working with Windows; My Computer & recycle bin; Desktop, Icons and Windows Explorer; Screen description & Working styles of Windows; Dialog Boxes & Toolbars; Working with Files & Folders; Shortcuts & Auto starts; Accessories and Windows setting using control Panel; Start button &	12	20

	Program List; Installing new Hardware's & Software's.		
II	Basic of MS Word: Creating Word Documents; The word Window, Entering Texts. Editing Document texts; Selecting Texts, Copying and moving Texts. Applying Text Enhancements; Applying fonts and fonts Styles in Word, Highlighting Text for Distinctive Look. Aligning and Formatting; Aligning Text using identification options, setting line spacing Options using Tabs. Cresting Lists, Number and symbols; Numbering and Bullets, Creating Special Characters, Replacing and checking Text; Creating and Applying Frequently used Texts, Finding and Replacing Texts, More about spelling and grammar using the thesaurus command, Getting Print using Print Preview, Changing Page orientation and paper size, Aligning Text Vertically, Setting Margins, Printing Options. Advanced Formatting Techniques in Word: Formatting Pages; Formatting Sections, Creating and Modifying Page Numbers, Creating Headers and Footers, Taking Care of Loose Ends, Working with Columns; Working with Newspaper Columns , Revising Columns structure. Constructing High- Quality Tables; Creating and Revising Tables, Modifying Table structure, Formatting Table, Creating Outlines in Word using Templates, Use of Mail Merge in Microsoft Word.	12	20
III	MS Access: Concepts & terms : database tables, relational database, records, fields, controls & objects, queries, forms, reports, properties, wizards, macros, MS Access requirements, Staring & quitting MS Access, MS Access Workspace, tool & views. Creating database & tables with & without wizard, field name, data types & Properties, adding & deleting fields, renaming fields & their caption, resizing fields, freezing columns, primary key field & indexing fields. MS Access Form; Form Wizard, saving & Modifying forms, Entering & editing data, finding, sorting & displaying data, creating queries, Using select queries and wild cards. MS Reports: Creating Reports, Previewing reports, Printing reports, Modifying & saving reports. Relational Databases: definition, purpose, creation, viewing, deleting. Expressions,	12	20

	Create Pivot Table or Pivot Chart Views in an Access desktop database.		
IV	Creating Excel Worksheets: Entering and Editing Cell Entries: Excel Application window, Workbooks and Worksheets, Moving the Cell Pointer, Entering Text and Numbers, Revising Text and Numbers. Working with Numbers; Creating Formulae, Formatting numbers. Changing worksheet Layout; Adjusting column width and Row Height, Inserting and Deleting Rows and Columns, Inserting and Deleting Cells, Moving and Copying Cell Contents, Naming Worksheets, Selecting Worksheets, Copying and Moving Worksheets, inserting and deleting Worksheets, other Formatting Options; Aligning Text, Border and colour. Printing in Excel; Print preview, Changing Page setup, Checking Worksheet Spelling. Advanced Techniques in Excel Using Functions and references: Use of functions, Entering Functions, Relative and Absolute Cell References. Create Named ranges, Creating Easy- to-understand Charts; Pie Charts, Series Charts, Creating Charts, Moving , Sizing and printing Chart Objects. Editing and Formatting Chart; Adding a Data series, Deleting a Data series, Modifying and Formatting Charts. Macros. Creation of Pivot table to analyse worksheet data.	12	20
V	Price determination under perfect competition and equilibrium of the firm, monopoly price and output determination and monopoly control, price determination under monopoly, imperfect and monopolistic competition- price determination. Creating a Basic Presentation, Building Presentations, Modifying Visual Elements, Formatting and Checking Text, Adding Objects, Applying Transitions, Animation Effects and Linking, Preparing handouts.	12	20

Unit – I

Introduction to MS Windows:

Microsoft Windows is a widely used operating system (OS) developed by Microsoft Corporation. It's designed to run on personal computers, laptops, tablets, and even servers. Since its initial release in 1985, Windows has evolved through numerous versions, each introducing new features and improvements. Here's a brief introduction to Windows along with some of its key features:

1. **Graphical User Interface (GUI):** One of the defining features of Windows is its graphical user interface, which allows users to interact with the computer using icons, windows, and menus rather than text-based commands.
2. **Multitasking:** Windows supports multitasking, allowing users to run multiple programs simultaneously. This feature enables users to switch between applications seamlessly.
3. **File Management:** Windows provides a file management system, including File Explorer, which allows users to organize, navigate, and manage files and folders stored on their computer or connected drives.
4. **Device Support:** Windows is compatible with a wide range of hardware devices, including printers, scanners, cameras, and external storage devices. Plug-and-play functionality makes it easy to connect and use these devices with Windows.
5. **Networking:** Windows includes built-in networking capabilities, allowing users to connect their computers to local area networks (LANs) or the internet. It supports various network protocols and services for sharing files, printers, and internet connections.
6. **Security Features:** Windows includes security features such as user accounts, access controls, and built-in antivirus software (Windows Defender) to help protect against malware, viruses, and other security threats.
7. **Software Compatibility:** Windows supports a vast ecosystem of software applications, ranging from productivity tools and multimedia software to games and development tools. Users can easily install and run third-party software on Windows.
8. **Updates and Support:** Microsoft regularly releases updates and patches to improve performance, fix bugs, and address security vulnerabilities in Windows. Additionally, Microsoft provides technical support and online resources to assist users with troubleshooting and problem-solving.
9. **Customization Options:** Windows offers various customization options, allowing users to personalize their desktop environment, including themes, wallpapers, and screen savers. Users can also customize system settings and preferences to suit their individual needs and preferences.
10. **Integration with Microsoft Services:** Windows integrates with various Microsoft services such as OneDrive (cloud storage), Office 365 (productivity suite), and Xbox Live (gaming platform), providing seamless access to these services from within the operating system.

Over the years, Microsoft has released numerous versions of the Windows operating system, each with its own set of features and targeted use cases. Here's an overview of some of the major versions of Windows and their typical use:

1. **Windows 3.x:** Released in the early 1990s, Windows 3.x was one of the first widely used versions of Windows. It introduced features like improved graphics support and support for multimedia applications. It was primarily used for basic productivity tasks and gaming.
2. **Windows 95/98/ME:** These versions of Windows were released in the mid-to-late 1990s and were aimed at home users and small businesses. They introduced significant improvements in user interface, networking capabilities, and multimedia support. They were widely used for personal computing tasks such as word processing, web browsing, and gaming.
3. **Windows NT/2000:** Windows NT and its successor Windows 2000 were designed for business and enterprise use. They introduced features like improved security, stability, and support for networking and server functionalities. They were commonly used in corporate environments for tasks such as file sharing, email, and server hosting.
4. **Windows XP:** Released in 2001, Windows XP was one of the most popular and widely used versions of Windows. It combined the stability of Windows NT/2000 with the consumer-friendly features of Windows 95/98. Windows XP was used for a wide range of tasks including personal computing, business use, and gaming.
5. **Windows Vista:** Released in 2007, Windows Vista introduced significant improvements in visual appearance, security, and system performance. However, it was criticized for its high system requirements and compatibility issues with certain hardware and software. It was used primarily by consumers and businesses who wanted the latest features and improvements.
6. **Windows 7:** Released in 2009, Windows 7 addressed many of the issues with Windows Vista and became one of the most popular versions of Windows. It introduced features like improved performance, better compatibility with hardware and software, and a streamlined user interface. Windows 7 was widely used in homes, businesses, and educational institutions.
7. **Windows 8/8.1:** Released in 2012, Windows 8 introduced a new touch-centric user interface designed for tablets and touchscreen devices. It also included improvements in performance, security, and cloud integration. However, the radical changes to the user interface were met with mixed reactions from users. Windows 8.1, released as a free update, addressed some of the criticisms of Windows 8.
8. **Windows 10:** Released in 2015, Windows 10 is the latest version of Windows and is designed to be a universal operating system for a wide range of devices including PCs, tablets, smartphones, and Xbox consoles. It combines the best features of Windows 7 and Windows 8/8.1 while introducing new features like virtual desktops, Cortana (virtual assistant), and the Windows Store. Windows 10 is widely used by consumers, businesses, and educational institutions around the world.

Working with Windows involves a variety of tasks, from basic navigation to more advanced operations. Here's a guide to help you work effectively with Windows:

1. **Basic Navigation:**
 - Start Menu: Click on the Start button to access programs, settings, and files.
 - Taskbar: The taskbar at the bottom of the screen displays open programs and allows you to switch between them.
 - Desktop: The desktop is where you can place shortcuts to programs and files for easy access.

2. File Management:

- File Explorer: Use File Explorer to navigate through files and folders on your computer.
- Copy, Move, and Delete: Right-click on files or folders to copy, move, or delete them.
- Organize: Use folders to organize your files and keep your computer tidy.

3. Customization:

- Personalize: Customize your desktop background, theme, and screen saver in the Personalization settings.
- Taskbar: Customize the taskbar by pinning frequently used programs and rearranging icons.

4. Applications:

- Installing: Install new applications from the Microsoft Store or by downloading them from the internet.
- Running: Open applications by clicking on their icons on the desktop or Start menu.
- Closing: Close applications by clicking on the "X" button in the top-right corner of the window.

5. Internet Browsing:

- Web Browser: Use Microsoft Edge or another web browser to browse the internet.
- Tabs: Open multiple web pages in separate tabs within the browser window.
- Favorites: Save frequently visited websites to your favorites for quick access.

6. Productivity Tools:

- Microsoft Office: Use programs like Word, Excel, and PowerPoint for word processing, spreadsheet creation, and presentations.
- Email: Access your email using the built-in Mail app or install a third-party email client.

7. Multimedia:

- Photos: Use the Photos app to view and organize your photos and videos.
- Music and Video: Use the Groove Music and Movies & TV apps to listen to music and watch movies.

8. Settings:

- Control Panel: Access the Control Panel to change system settings such as display resolution, power options, and device management.
- Settings App: Use the Settings app to manage Windows settings such as network, accounts, and privacy.

9. Security:

- Windows Defender: Enable and configure Windows Defender to protect your computer from viruses and malware.
- Updates: Keep your computer up to date by installing Windows updates regularly to ensure security and performance improvements.

10. Help and Support:

- Help Menu: Access the Windows Help and Support menu for assistance with troubleshooting and finding solutions to common problems.
- Online Resources: Visit the Microsoft website or community forums for additional help and support.

By mastering these basic skills and familiarizing yourself with the features and functions of Windows, you can work more efficiently and effectively with your computer.

Unit II

Basic Concept of MS. Word :-

Microsoft Word is a powerful word processing application that allows you to create, edit, format, and share documents. Here's a basic overview of some of the key features and functions of MS Word:

1. Creating a New Document:

- Open MS Word and click on "Blank document" to start a new document.
- Alternatively, you can choose from various templates for different types of documents like resumes, letters, reports, etc.

2. Entering and Editing Text:

- Click anywhere in the document to start typing.
- Use the keyboard to enter text. You can edit text by using the arrow keys, backspace, delete, or by clicking and dragging to select text.

3. Formatting Text:

- Use the formatting options on the toolbar to change the font, font size, color, and style of your text.
- You can also apply formatting such as bold, italic, underline, and strikethrough to emphasize or highlight text.

4. Paragraph Formatting:

- Use the alignment options (left, center, right, justify) to control the alignment of your text.
- Adjust line spacing, indentation, and paragraph spacing to format your paragraphs.

5. Inserting Objects:

- Insert images, shapes, charts, and other objects into your document by clicking on the "Insert" tab.
- You can also insert tables, links, headers, footers, and page numbers to enhance your document.

6. Page Layout:

- Customize the page layout by adjusting margins, orientation (portrait or landscape), and paper size.
- You can also add page breaks and adjust the layout of columns in your document.

7. Spell Check and Proofing:

- Use the built-in spell checker to correct spelling errors and thesaurus to find synonyms for words.
- Review your document for grammar and style issues using the proofing tools.

8. Saving and Sharing:

- Save your document by clicking on the "Save" button or pressing Ctrl + S. Choose a location and file name for your document.
- You can share your document by sending it as an email attachment, sharing it via cloud storage services like OneDrive, or by printing it.

9. Printing:

- Click on the "File" tab, select "Print," and choose your printing options such as printer, page range, and number of copies.
- Preview your document before printing to ensure it looks the way you want.

10. Keyboard Shortcuts:

- Learn keyboard shortcuts to perform common tasks quickly. For example, Ctrl + B for bold, Ctrl + I for italic, Ctrl + U for underline, etc.

By mastering these basic features and functions of MS Word, you'll be able to create professional-looking documents for various purposes such as reports, letters, resumes, and more.

Creating Word documents is a straightforward process in Microsoft Word. Here's a step-by-step guide:

1. **Open Microsoft Word:** Launch Microsoft Word by clicking on its icon in the Start menu (Windows) or in the Applications folder (Mac). You can also search for "Word" in the search bar to find and open it.
2. **Start a New Document:**
 - Once Word is open, you'll typically see a blank document ready for you to start typing.
 - If you want to start with a specific template, you can choose one from the template gallery. Click on "File" > "New" and then select a template that suits your needs.
3. **Enter Text:**
 - Click anywhere in the document to place the cursor, and start typing to enter your text.
 - You can type freely or copy and paste text from other sources like a web page or another document.
4. **Formatting Text:**
 - To format text, select the text you want to format by clicking and dragging your cursor over it.
 - Use the formatting options in the toolbar or ribbon at the top of the window to change the font, font size, style (bold, italic, underline), color, and alignment of the selected text.
5. **Inserting Objects:**
 - You can insert various objects into your document such as images, shapes, charts, tables, and SmartArt.
 - Click on the "Insert" tab in the toolbar and choose the type of object you want to insert.
6. **Page Layout:**
 - Customize the layout of your document by adjusting margins, orientation (portrait or landscape), and paper size.
 - You can also add page breaks, adjust the layout of columns, and set up headers and footers.
7. **Saving Your Document:**
 - Save your document by clicking on the "File" tab and selecting "Save" or pressing Ctrl + S (Cmd + S on Mac).
 - Choose a location on your computer to save the file, give it a name, and select the desired file format (usually .docx for Word documents).
8. **Reviewing and Editing:**
 - Use the spell checker and grammar checker to review your document for errors.
 - Proofread your document for any mistakes or inconsistencies and make necessary edits.

9. Sharing and Collaboration:

- Once your document is ready, you can share it with others by sending it via email, sharing it through cloud storage services like OneDrive, or by printing it.
- If you're collaborating with others on the document, you can use Word's built-in collaboration features like comments and track changes to work together more effectively.

10. Printing:

- To print your document, click on the "File" tab, select "Print," and choose your printing options such as printer, page range, and number of copies.
- Preview your document before printing to ensure it looks the way you want.

That's it! By following these steps, you can create professional-looking Word documents for various purposes such as reports, letters, resumes, and more.

Formatting of Pages in MS Word:-

Formatting pages in Microsoft Word allows you to customize the layout, appearance, and structure of your document. Here are some common formatting options for pages in MS Word:

1. Page Size and Orientation:

- Click on the "Layout" tab in the ribbon.
- Use the "Size" dropdown menu to choose a predefined page size or select "More Paper Sizes" to customize the size.
- Use the "Orientation" dropdown menu to switch between portrait and landscape orientation.

2. Margins:

- Click on the "Layout" tab in the ribbon.
- Click on the "Margins" dropdown menu to choose predefined margin settings (Normal, Narrow, Wide, etc.) or select "Custom Margins" to set your own margin measurements.

3. Page Breaks:

- To insert a manual page break, place the cursor where you want to start a new page.
- Go to the "Insert" tab in the ribbon and click on "Page Break" in the Pages group.

4. Headers and Footers:

- Click on the "Insert" tab in the ribbon.
- Click on "Header" or "Footer" in the Header & Footer group, then select a built-in option or choose "Edit Header" or "Edit Footer" to create a custom header or footer.
- Insert page numbers, document titles, dates, and other information into the header or footer.

5. Page Numbers:

- Click on the "Insert" tab in the ribbon.
- Click on "Page Number" in the Header & Footer group and select where you want the page numbers to appear (Top of Page, Bottom of Page, Page Margins, etc.).
- Choose a predefined page number format or select "Format Page Numbers" to customize the numbering style.

6. Columns:

- Click on the "Layout" tab in the ribbon.
- Click on "Columns" in the Page Setup group and select the number of columns you want to divide your document into.

7. Backgrounds:

- To add a background color or image to your document, go to the "Design" tab in the ribbon.
- Click on "Page Color" in the Page Background group and choose a color, or select "Fill Effects" to add a gradient, texture, or pattern.
- You can also insert a watermark by clicking on "Watermark" and selecting a built-in option or customizing your own.

8. Page Borders:

- Click on the "Design" tab in the ribbon.
- Click on "Page Borders" in the Page Background group and choose a border style, color, width, and art style for your page borders.

By using these formatting options, you can customize the appearance and layout of your Word documents to suit your needs and preferences.

Unit – III

MS Access: Concepts & terms:-

Microsoft Access is a powerful database management system that allows users to create and manage databases. Here are some key concepts and terms related to MS Access:

1. **Database:** A database is a collection of related data organized in a structured format. In MS Access, a database typically consists of tables, queries, forms, reports, macros, and modules.
2. **Table:** A table is a collection of related data organized in rows (records) and columns (fields). Each column represents a different attribute or field, and each row represents a single record. Tables are used to store and organize data in MS Access.
3. **Field:** A field is a single piece of information within a table. Each field corresponds to a specific attribute of the data being stored, such as a person's name, age, or address. Fields have data types (e.g., text, number, date/time) that define the type of data they can store.
4. **Record:** A record is a complete set of related fields within a table. Each record represents a single entity or item in the database, such as a customer, product, or order.
5. **Primary Key:** A primary key is a unique identifier for each record in a table. It ensures that each record can be uniquely identified and distinguishes one record from another. Primary keys are typically used to establish relationships between tables.
6. **Query:** A query is a request for information from a database. Queries are used to retrieve, filter, and manipulate data stored in tables based on specified criteria. MS Access provides a graphical interface for creating queries using a query design view.
7. **Form:** A form is a user interface that allows users to interact with data in a database. Forms are used to enter, edit, and view data in a more user-friendly and structured way than directly working with tables. Forms can be customized to display specific fields and control user input.
8. **Report:** A report is a formatted presentation of data from a database. Reports are used to summarize, analyze, and present data in a visually appealing and meaningful format. MS Access provides tools for creating and customizing reports to meet specific reporting requirements.
9. **Macro:** A macro is a predefined set of actions or commands that automate tasks in MS Access. Macros can be used to automate repetitive tasks, such as opening forms, running queries, or printing reports, without requiring programming knowledge.
10. **Module:** A module is a container for Visual Basic for Applications (VBA) code in MS Access. Modules allow users to write custom code to extend the functionality of Access beyond what is possible with macros. VBA code can be used to create custom functions, automate tasks, and add advanced features to databases.

Understanding these concepts and terms is essential for working effectively with Microsoft Access and building databases to manage and analyze data.

Creating a Table and Database:-

To create a database and tables in Microsoft Access, follow these steps:

1. **Open Microsoft Access:** Launch Microsoft Access on your computer.
2. **Create a New Database:**
 - Click on the "Blank Database" option when prompted to create a new database.

- Alternatively, you can choose "File" > "New" > "Blank Database" from the menu.
- 3. **Save the Database:**
 - Choose a location on your computer to save the database file.
 - Enter a name for the database file and click "Create" or "OK" to save it.
- 4. **Design Tables:**
 - Once the new database is created, you'll see a blank screen with the name of your database at the top.
 - Click on the "Table Design" button in the "Tables" group on the ribbon to start designing a new table.
- 5. **Add Fields:**
 - In the table design view, you'll see a grid with columns for field names, data types, and other properties.
 - Enter the names of the fields you want to include in your table in the "Field Name" column.
 - Select the appropriate data type for each field from the dropdown menu in the "Data Type" column (e.g., Text, Number, Date/Time).
 - You can also specify additional properties for each field, such as field size, format, and validation rules.
- 6. **Set Primary Key:**
 - Select the field that will serve as the primary key for your table by clicking on its row selector (the gray box to the left of the field name).
 - Click on the "Primary Key" button in the "Tools" group on the ribbon to set the selected field as the primary key.
- 7. **Save the Table:**
 - Click on the "Save" button in the Quick Access Toolbar or press Ctrl + S to save the table.
 - Enter a name for the table and click "OK" to save it.
- 8. **Repeat for Additional Tables:**
 - If your database requires multiple tables, repeat the process to create additional tables by clicking on the "Table Design" button and designing each table accordingly.
- 9. **Establish Relationships (Optional):**
 - After creating tables, you can establish relationships between them to link related data.
 - Click on the "Database Tools" tab, then click on "Relationships" to open the Relationships window.
 - Drag and drop the primary key field from one table to the matching foreign key field in another table to create a relationship.
- 10. **Save the Database:**
 - Once you've created and designed all the necessary tables, save the database by clicking on the "Save" button in the Quick Access Toolbar or pressing Ctrl + S.

That's it! You've successfully created a database and tables in Microsoft Access. You can now start entering data into your tables and using them to manage and analyze your information.

MS Access Form

Microsoft Access forms provide a user-friendly interface for interacting with data in a database. Here's how to create a form in MS Access:

1. **Open Your Database:** Launch Microsoft Access and open the database in which you want to create the form.
2. **Navigate to Forms:** In the left navigation pane, click on the "Forms" tab. This will display any existing forms in your database.
3. **Create a New Form:**
 - Click on the "Create" tab in the ribbon.
 - Select "Form Design" from the Forms group to open the form design view.
4. **Add Fields to the Form:**
 - In the form design view, you'll see a blank canvas where you can design your form.
 - To add fields from a table or query to your form, click on the "Add Existing Fields" button in the Tools group on the ribbon.
 - This will open the "Field List" pane, which displays all the fields from the selected table or query.
 - Drag and drop fields from the Field List pane onto the form canvas to add them to the form.
5. **Arrange and Customize Fields:**
 - Once you've added fields to the form, you can arrange and customize them as needed.
 - Click and drag fields to reposition them on the form.
 - Resize fields by clicking and dragging the sizing handles.
 - Customize field properties such as labels, captions, and formatting by right-clicking on the field and selecting "Properties."
6. **Add Controls and Labels:**
 - In addition to fields, you can also add other controls to your form, such as buttons, text boxes, combo boxes, and checkboxes.
 - Click on the "Design" tab in the ribbon, then select the control you want to add from the Controls group.
 - Click on the form canvas to place the control, then resize and customize it as needed.
7. **Save and Preview the Form:**
 - Once you've designed your form, save it by clicking on the "Save" button in the Quick Access Toolbar or pressing Ctrl + S.
 - Enter a name for the form and click "OK" to save it.
 - To preview the form, switch to the Form View by clicking on the "View" button in the Views group on the ribbon.
8. **Make Further Adjustments:**
 - After previewing the form, you may need to make further adjustments to the layout or design.
 - Switch back to the Design View by clicking on the "View" button and selecting "Design View" from the dropdown menu.
 - Continue refining the form until you're satisfied with the design.
9. **Close and Use the Form:**
 - Once you're done designing the form, close the form design view by clicking on the "Close" button in the upper-right corner of the form window.
 - Your form will now be available in the Forms tab of your database, where you can open and use it to interact with your data.

By following these steps, you can create a custom form in Microsoft Access to input, edit, and view data in your database.

MS Reports:-